Hanqi Liu

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SUMMARY

- Seeking an intern software engineer position starting from June 2026, Computer Science student with 3 years of project experience in various areas, including web development, machine learning, mobile development. Strong knowledge of computer science, mathematics and statistics, and solid programming skills in Java, MySQL, JavaScript, Python, hands-on experience with Spring, AWS, React, Node.js, Ant Design.

EDUCATION

New York University (Tandon School of Engineering) *Brooklyn NY* **Sep. 2023 - May 2027**

B.S. Mathematics and Computer Science, GPA 3.5/4.0

- Relevant coursework: Algorithms & Data Structures, Database Design, Object-Oriented Programming, Operating Systems, Discrete Mathematics

TECHNICAL SKILL

- Languages: Python, Java, JavaScript, C++, C, R
- Frameworks & Technologies: React, Node.js, Spring Boot, Express.js, Android Jetpack, LangChain
- Databases & Cloud: MySQL, PostgreSQL, AWS (RDS, ECR, App Runner), Google Cloud (App Engine, GCE)
- Tools: Git, Docker, ElasticSearch, QEMU/WSL2, JWT, REST APIs

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern | Xiaomi (largest smartphone company in China) | Beijing, China **Jun – Aug 2024**

- Contributed to the development of Xiaomi Auto's in-vehicle voice assistant under Xiaomi HyperOS, implementing a complete speech pipeline covering wake word, **ASR**, **NLU**, and **TTS**.
- Designed and built a microservice-based architecture using **FastAPI** + **gRPC**, enabling independent deployment and real-time communication between voice modules.
- Implemented an **offline-first caching** mechanism that locally stores top user queries (e.g., navigation, media, A/C control), reducing average response latency by 35% in weak-connectivity environments.

Software Engineer Intern | PLCT Lab | Beijing, China **Jun – Aug 2025**

- Collaborated with the PLCT infrastructure team to port and validate the OnlineOrder full-stack system across heterogeneous environments, including **AWS** App Runner (x86_64) and PLCT's **RISC-V** virtual clusters.
- Built multi-architecture **Docker** images (x86_64 + riscv64) using Docker Buildx and published them to **AWS ECR**, enabling unified deployment and testing workflows.
- Optimized backend performance of the **Spring Boot** + **PostgreSQL** services, reducing API latency by 30% and improving throughput by 40% under simulated 1,000-user loads.
- Documented reproducible containerization procedures demonstrating cross-platform portability between AWS and RISC-V environments.

PROJECTS

SocialAI: AI-Enhanced Social Network

- Launched a **Go**-based web service on **Google App Engine** that handled post creation, feed generation, and user interactions, efficiently scaling to support over 10,000 simulated posts.
- Integrated **OpenAI's DALL-E 3 API**, allowing users to generate custom images for their posts, which increased user engagement metrics (posts with images) by 60%.
- Implemented a **JWT**-based authentication flow with **React Router**, securing user sessions and reducing unauthorized API access incidents to zero.
- Utilized **Elasticsearch** deployed on **Google Compute Engine** to power real-time search, enabling users to find relevant posts from a dataset of 50,000+ entries with sub-second latency.

MiniSpotify: Android Music Application

- Developed a native **Android** app using **Kotlin** and **Jetpack Compose**, following **MVVM** architecture with **Hilt** for dependency injection, which decreased class coupling and improved unit testing coverage by 30%.
- Integrated a mock **RESTful API** using **JSON-Server** and **Retrofit**, simulating a production environment with a library of 100+ songs and enabling seamless data fetching for feeds, albums, and user favorites.
- Implemented offline caching for favorite songs using **Room Database**, which reduced redundant network calls by 40% and provided a seamless user experience in low-connectivity scenarios.
- Incorporated **ExoPlayer** for global audio playback management, ensuring smooth, uninterrupted music streaming across different app screens and supporting background playback.

Agent AI: Document Q&A System with Voice Interface

- Engineered a **Retrieval-Augmented Generation** (RAG) system using **GPT-4** and **LangChain**, processing a corpus of 500+ PDF documents to achieve a 92% accuracy rate on factual question-answering tasks.
- Designed a hybrid search strategy combining **vector similarity search** with keyword matching via **Elasticsearch**, improving answer relevance by 35% compared to a baseline keyword-only approach.
- Implemented a voice-enabled **React** frontend using **react-speech-recognition**, reducing the average time for users to submit queries by 50% compared to manual typing.
- Deployed a scalable **Node.js** and **Express** backend on **Google App Engine**, orchestrating multiple APIs and maintaining an average response time under 2 seconds for complex document analysis requests. Agent AI: Full - Stack RAG & MCP - Based Document Q&A Agent.